

Optional Proofs — Progress Update

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Lighthouse Implementation

- ☒ Proof engine service
- ☒ Execution proof gossip
- ☒ Execution proof signature verification
- ☒ Validator proof service
- ☒ Proof engine <-> Beacon chain integration
- ☒ Unit tests
- ☒ Network proof sync protocol (naive)
- ☒ Integration testing
-  ExecutionProofStatus RPC — PR [feat/execution-status](#)
-  ExecutionProofStatus based network sync - PR [feat/execution-status](#)
- ☐ Validator proof re-signing
- ☐ Ban validators for invalid proofs
- ☐ Kurtosis support

ExecutionProofStatus RPC

Problem:

- Only the peer initiating the connection has the remote peer's ENR - cannot determine proof capabilities symmetrically
- No awareness of proof availability on the peer

Solution:

- `ExecutionProofStatus` invoked eagerly on connection to peers advertising `eproof` in ENR
- Exchanges `{ slot, block_root }` bidirectionally
- Both peers learn each other's latest verified proof state and proof capabilities

Benefits:

- Select best peer to sync from based on execution proof status
- Decide when to do **range sync** vs **by-root sync**
- No changes to existing gossip protocol, purely additive